

PEM catalyst layer thickness Code Revised

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Problem Description & Metrics

Problem:

- Need to quantitatively analyze the morphology of pre-segmented catalyst layer shapes, which may be complex, curved, or non-uniform.

Challenge:

- Requires robust geometric measurements that go beyond simple manual or columnar analysis, including local thickness at every point and the "twistedness" (tortuosity) of the primary path.

Goal:

- Develop a robust, automated method using computer vision to extract meaningful morphological metrics from isolated shapes.

Key Metrics Calculated:

Shape & Geometric Metrics:

- **Area:** The total area of the shape.
- **Perimeter:** The length of the shape's outer boundary.
- **Circularity:** A measure of roundness (1.0 = perfect circle).
- **Solidity:** A measure of convexity, comparing the shape's area to the area of its convex hull.
- **Tortuosity:** The ratio of the actual path length of the shape's primary axis to the straight-line distance between its endpoints.

Thickness Metrics (from Medial Axis):

- Mean, Median, Min, Max, and Standard Deviation of the local thickness, measured accurately along the central skeleton of the shape.

Internal Structure/Texture:

- **Internal Intensity:** Statistical moments (Mean, Std Dev, Skewness, Kurtosis) of the grayscale pixel values within the original mask.

Analysis Approach & Methodology

Automated Image Analysis Pipeline:

1. Load, Scale, & Isolate Mask:

- Loads the input image, which contains a pre-masked, grayscale shape on a white background.
- Calculates pixel size ($\mu\text{m}/\text{pixel}$) using the Horizontal Field Width (HFW) from image metadata.
- Automatically isolates the non-white shape to create a high-fidelity binary mask, replacing the previous multi-step segmentation process.

2. Core Morphological Analysis:

- A **Euclidean Distance Transform** is calculated on the binary mask. This determines the shortest distance from every pixel inside the shape to the nearest boundary, creating a "local radius" map.
- **Skeletonization** is used to compute the medial axis (the central skeleton) of the shape.

3. Primary Axis & Tortuosity Calculation:

- The code identifies the primary axis by finding the two most distant endpoints on the skeleton.
- The **A* search algorithm** finds the true, shortest path along the skeleton pixels between these two main endpoints.
- **Tortuosity** is then calculated as the ratio of this true path length to the straight-line distance between the endpoints.

4. Metric Calculation:

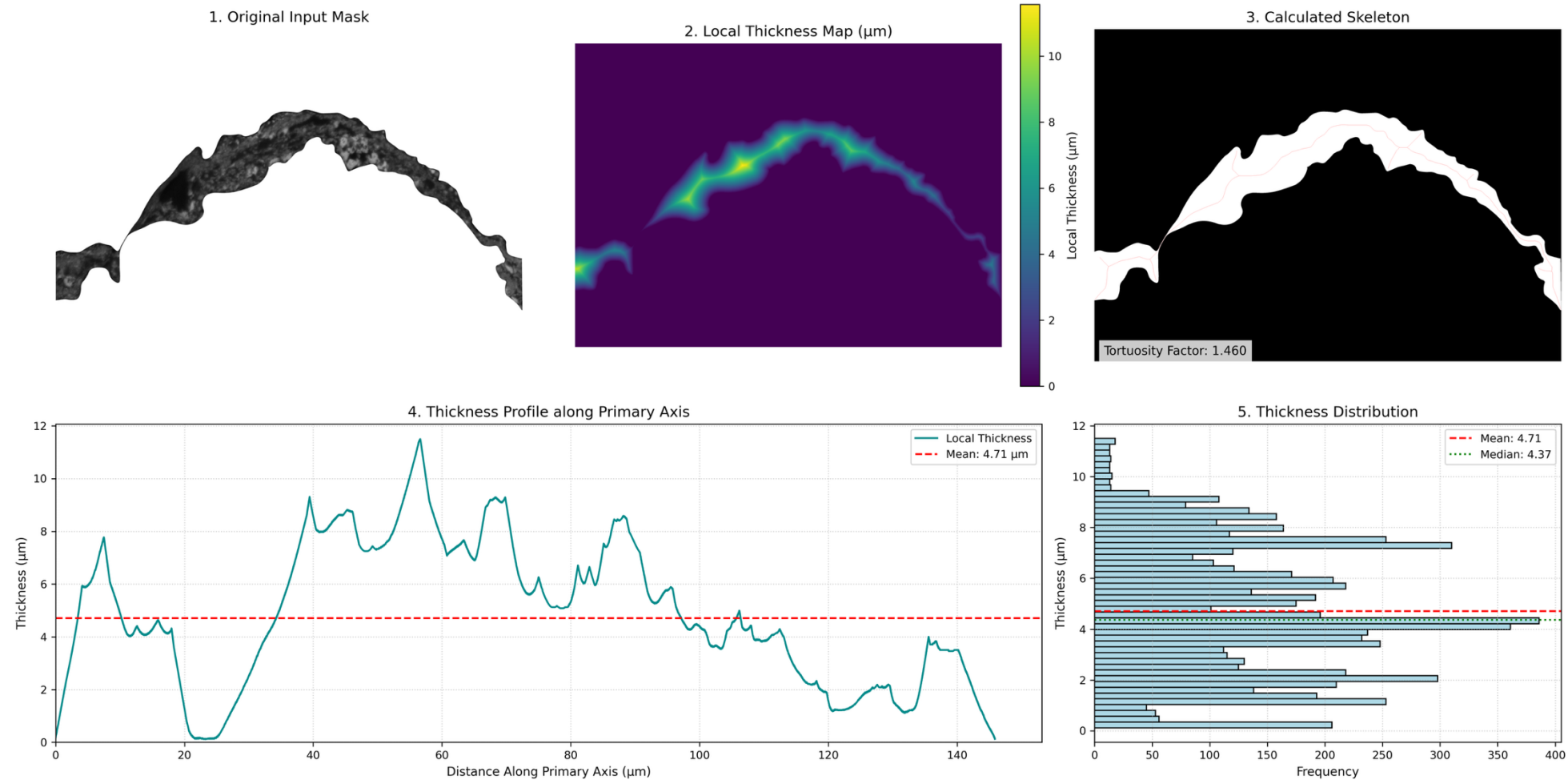
- Computes a full suite of metrics based on the analysis.
- **Thickness statistics** are derived from the skeleton and the distance transform map.
- **Shape metrics** (Area, Perimeter, Circularity, Solidity) and **Tortuosity** are calculated from the mask's contour and the primary axis path.
- **Internal intensity statistics** are calculated from the original grayscale values within the isolated mask.

5. Visualization & Output:

- Generates a new 5-panel composite figure showing:
 - Original Input Mask
 - Color-Coded Local Thickness Map
 - Calculated Skeleton (annotated with Tortuosity Factor)
 - Thickness Profile vs. Distance Along Primary Axis
 - Thickness Distribution Histogram
- Provides the option to save the composite figure and the calculated metrics to separate files.
- Includes the new capability to export each of the five panels as an individual, publication-ready image file.

Analysis Results

Morphological Analysis: example1.png



Analysis Results

Morphological Analysis: example2.png

